

RB-SQMC with Kronecker Structure

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1 Setup

We define a correlated OU-process in $\mathbb{R}^F$. Let $X_t \in \mathbb{R}^{P \times F}$ be the latent states at time t . The initial distribution is a multivariate Gaussian in $\mathbb{R}^{P \times F}$ with a Kronecker product structure,

$$X_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$$

where $\Sigma_0 = \Gamma_0 \otimes B \in \mathbb{R}^{PF \times PF}$ is a Kronecker product of $\Gamma_0 \in \mathbb{R}^{F \times F}$ and $B \in \mathbb{R}^{P \times P}$, where Γ_0 is a time-varying covariance matrix and B is a static covariance matrix within a index $i \in \{1, \dots, P\}$

The state transition is governed by the OU-process defined as

$$X_t = \mu_0 + \Phi_t(X_{t-1} - \mu_0) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t)$$

where $Q_t = \Sigma_0 - \Phi_t \Sigma_0 \Phi_t^\top$ and $\Phi_t = \phi_t \otimes I_P$ with $\phi_t = \exp(-\kappa \Delta t) \cdot I_F \in \mathbb{R}^{F \times F}$, where $Q_t \in \mathbb{R}^{PF \times PF}$ is also a non-diagonal covariance matrix.

Observations arrive at discrete time steps $t = 1, \dots, T$. Let $y_t \in \mathbb{R}^H$ be the observed data at time t where $H \leq F$ and $X_t^E \in \mathbb{R}^{H \times P}$ be the latent states involved in an observation. The likelihood function is defined as a non-linear function of inputs X_t^E ,

$$G_t(y_t | X_t^E)$$

$X_t^R \in \mathbb{R}^{(F-H) \times P}$ are the latent states not involved in an observation. This notation will be used in future sections.

2 Rao-Blackwellized Particle Filter (RB-PF)

The RB-PF targets the marginal distribution of latent states $X_t^E \subseteq X_t$ involved in an observation y_t . The joint distribution of latent states and observed can be factorized as

$$p(X_{0:T} | y_{1:T}) = p(X_{0:T}^{-E} | X_{0:T}^E) p(X_{0:T}^E | y_{1:T})$$

where $X_{0:T}^{-E}$ are the latent states not involved in an observation. Since the dynamics of latent states are linear and Gaussian, the conditional distribution of $X_{0:T}^{-E} | X_{1:T}^E$ is deterministic and computed exactly using the Kalman filter.

Bootstrap Particle Filter

The particle filter performs the weight update as

$$\tilde{w}_t \propto w_{t-1} \cdot \frac{G_t(y_t | X_t) p(X_t | X_{t-1})}{q(X_t | X_{t-1})}$$

where $\tilde{w}_t$ is the unnormalized weight at time t . Under the bootstrap particle filter, the proposal distribution is the transition distribution $q(X_t | X_{t-1}) = p(X_t | X_{t-1})$. The weight update simplifies to

$$\tilde{w}_t \propto w_{t-1} \cdot G_t(y_t | X_t)$$

$$\log(\tilde{w}_t) = \log(w_{t-1}) + \log G_t(y_t | X_t)$$

Algorithm

0 Initialization: Our initial distribution follows a multivariate Gaussian with Kronecker structure. We generate N particles with equal weights $w_0^{(i)} = 1/N$ for $i = \{1, \dots, N\}$.

$$X_0^{(i)} \sim \mathcal{N}(\mu_0, \Sigma_0)$$

where $\Sigma_0 = \Gamma_0 \otimes B$ is a Kronecker product of $\Gamma_0 \in \mathbb{R}^{F \times F}$ and $B \in \mathbb{R}^{P \times P}$, where Γ_0 is a time-varying covariance matrix and B is a static covariance matrix.

2 Prediction: Compute the predictive mean and covariance for the full state $X_t \mid y_{1:t-1}$ using the Kalman filter equations since it is linear and Gaussian.

$$\mu_{t|t-1}^{(i)} = \mu_0 + \Phi_t(\mu_{t-1}^{(i)} - \mu_0)$$

$$P_{t|t-1}^{(i)} = \Phi_t P_{t-1}^{(i)} \Phi_t^\top + Q_t = \Phi_t \Sigma_{t-1}^{(i)} \Phi_t^\top + Q_t$$

3 Exact Gaussian Marginalization: The predictive distribution for the latent states involved in an observation $X_t^{E,(i)} \mid y_{1:t-1}$ is a multivariate Gaussian conditional on the observed data $y_{1:t-1}$.

$$\begin{pmatrix} X_t^{E,(i)} \\ X_t^{R,(i)} \end{pmatrix} \mid y_{1:t-1} \sim \mathcal{N} \left(\begin{bmatrix} \mu_t^{E,(i)} \\ \mu_t^{R,(i)} \end{bmatrix}, \begin{bmatrix} P_t^{EE,(i)} & P_t^{ER,(i)} \\ P_t^{RE,(i)} & P_t^{RR,(i)} \end{bmatrix} \right)$$

- $P_t^{EE,(i)} = \text{Var}(X_t^{E,(i)} \mid y_{1:t-1}) \in \mathbb{R}^{HP \times HP}$
- $P_t^{RR,(i)} = \text{Var}(X_t^{R,(i)} \mid y_{1:t-1}) \in \mathbb{R}^{(F-H)P \times (F-H)P}$
- $P_t^{ER,(i)} = \text{Cov}(X_t^{E,(i)}, X_t^{R,(i)} \mid y_{1:t-1}) = (P_t^{RE,(i)})^\top \in \mathbb{R}^{HP \times (F-H)P}$ since the covariance matrix is symmetric.

4 Bootstrap particle sampling: Extract the predictive mean and covariance for the latent states involved in an observation $X_t^{E,(i)} \mid y_{1:t-1}$.

$$X_t^{E,(i)} \mid y_{1:t-1} \sim \mathcal{N}(\mu_t^{E,(i)}, P_t^{EE,(i)})$$

where

$$\mu_t^{E,(i)} = \begin{bmatrix} \mu_{t|t-1}^{(i)}[f_1] \\ \mu_{t|t-1}^{(i)}[f_2] \\ \vdots \\ \mu_{t|t-1}^{(i)}[f_H] \end{bmatrix} \in \mathbb{R}^{HP}$$

$$P_t^{EE,(i)} = \begin{bmatrix} P_{t|t-1}^{(i)}[f_1, f_1] & P_{t|t-1}^{(i)}[f_1, f_2] & \cdots & P_{t|t-1}^{(i)}[f_1, f_H] \\ P_{t|t-1}^{(i)}[f_2, f_1] & P_{t|t-1}^{(i)}[f_2, f_2] & \cdots & P_{t|t-1}^{(i)}[f_2, f_H] \\ \vdots & \vdots & \ddots & \vdots \\ P_{t|t-1}^{(i)}[f_H, f_1] & P_{t|t-1}^{(i)}[f_H, f_2] & \cdots & P_{t|t-1}^{(i)}[f_H, f_H] \end{bmatrix} \in \mathbb{R}^{HP \times HP}$$

Sample the HP -dimensional latent states involved in an observation $X_t^{E,(i)} \mid y_{1:t-1}$ for each particle $i = 1, \dots, N$.

5 Weight Update: Compute the unnormalized weights $\tilde{w}_t^{(i)}$ for each particle $i = 1, \dots, N$ using the likelihood function $G_t(y_t \mid X_t^{E,(i)})$.

$$\log(\tilde{w}_t^{(i)}) = \log(w_{t-1}^{(i)}) + \log G_t(y_t \mid X_t^{E,(i)})$$

6 Kalman Update: By treating the sampled latent states $X_t^{E,(i)} = \hat{\mu}_t^{E,(i)}$ as noise-free measurements, we can then apply the standard Kalman update to the remaining latent states $X_t^{R,(i)} \mid X_t^{E,(i)} = \hat{\mu}_t^{E,(i)}, y_{1:t-1}$ for each particle $i = 1, \dots, N$.

$$X_t^{R,(i)} \mid X_t^{E,(i)} = \hat{\mu}_t^{E,(i)}, y_{1:t-1} \sim \mathcal{N}(\mu_t^{R|E,(i)}, P_t^{RR|E})$$

where

$$\mu_t^{R|E,(i)} = \mu_{t|t-1}^{R,(i)} + K_t^{(i)}(\hat{\mu}_t^{E,(i)} - \mu_t^{E,(i)})$$

$$P_t^{RR|E,(i)} = P_t^{RR,(i)} - K_t^{(i)} P_{t|t-1}^{ER,(i)}$$

The exact Kalman Gain equation is given by decomposing the predictive covariance following the Kronecker product structure:

$$\begin{aligned} K_t^{(i)} &= P_{t|t-1}^{RE,(i)} (P_{t|t-1}^{EE,(i)})^{-1} \\ &= (\Gamma_t^{RE,-} \otimes B) (\Gamma_t^{EE,-} \otimes B)^{-1} \\ &= (\Gamma_t^{RE,-} \otimes B) ((\Gamma_t^{EE,-})^{-1} \otimes B^{-1}) \\ &= \Gamma_t^{RE,-} (\Gamma_t^{EE,-})^{-1} \otimes I_P \end{aligned}$$

where $\Gamma_t^{EE,-} \in \mathbb{R}^{H \times H}$ and $\Gamma_t^{RE,-} \in \mathbb{R}^{(F-H) \times H}$ are the submatrices of Γ_t^- satisfying $P_{t|t-1}^{EE,(i)} = \Gamma_t^{EE,-} \otimes B$ and $P_{t|t-1}^{RE,(i)} = \Gamma_t^{RE,-} \otimes B$.

The computational complexity of the Kalman update is $O(H^3)$ from the inversion of $\Gamma_t^{EE,-} \in \mathbb{R}^{H \times H}$ and $O(H^2 + (F-H)H)$ from storage of the covariance matrices.

The Kalman update is then applied for each particle $i = 1, \dots, N$ to obtain the updated mean and covariance for the remaining latent states $X_t^{R,(i)} \mid X_t^{E,(i)} = \hat{\mu}_t^{E,(i)}, y_{1:t-1}$.

7 Resampling: Resample particles $X_t^{E,(i)}$ based on $\tilde{w}_t^{(i)}$ and set $w_t^{(i)} = 1/N$ for $i = 1, \dots, N$.

3 RB-PF Smoothing

We can apply the Forward Filtering Backwards Sampling (FFBSi) to the Rao-Blackwellized particles. The forward filter approximates the distribution at time T as a Gaussian mixture as stated previously.

$$p(X_T \mid y_{1:T}) \approx \sum_{i=1}^N w_T^{(i)} \mathcal{N} \left(\begin{pmatrix} \mu_T^{E,(i)} \\ m_T^{R|E,(i)} \end{pmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & P_T^{RR|E} \end{bmatrix} \right)$$

Algorithm

1 Initialization: To start the backward trajectory, we first **select which mixture component** (particle) to sample from, then **sample the actual state** from that component:

1. Sample component index: $I_T \sim \text{Categorical}(w_T^{(1)}, \dots, w_T^{(N)})$

2. Sample state from component I_T :

$$X_T^* = \begin{pmatrix} X_T^{E,*} \\ X_T^{R,*} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_T^{E,(I_T)} \\ m_T^{R|E,(I_T)} \end{pmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & P_T^{RR|E} \end{bmatrix} \right)$$

Since $X_T^{E,(I_T)}$ has zero covariance, $X_T^{E,*} = \mu_T^{E,(I_T)}$ exactly (degenerate/point mass). Only $X_T^{R,*}$ is randomly sampled from $\mathcal{N}(m_T^{R|E,(I_T)}, P_T^{RR|E})$. This gives us one complete smoothed state X_T^* to begin the backward pass.

2 Backward Simulation of X_t^* : For $t = T - 1 \dots 0$, we sample X_t^* from the conditional distribution $p(X_t | X_{t+1}^*, y_{1:t})$.

$$\begin{aligned} p(X_t | X_{t+1}^*, y_{1:t}) &\propto p(X_{t+1}^* | X_t) p(X_t | y_{1:t}) \\ &= p(X_{t+1}^* | X_t) \sum_{i=1}^N w_t^{(i)} \mathcal{N}(X_t | \mu_t^{(i)}, \Sigma_t^{(i)}) \end{aligned}$$

Since $p(X_{t+1}^* | X_t) = \mathcal{N}(X_{t+1}^* | \mu_0 + \Phi_{t+1}(X_t - \mu_0), Q_{t+1})$ is linear Gaussian, the product yields a new Gaussian mixture

$$p(X_t | X_{t+1}^*, y_{1:t}) \approx \sum_{i=1}^N w_{t|t+1}^{(i)} \mathcal{N}(X_t | m_{t|t+1}^{(i)}, P_{t|t+1}^{(i)})$$

The backward weights are computed as

$$w_{t|t+1}^{(i)} \propto w_t^{(i)} \cdot \mathcal{N}(X_{t+1}^* | \mu_{t+1|t}^{(i)}, P_{t+1|t}^{(i)})$$

where

$$\mu_{t+1|t}^{(i)} = \mu_0 + \Phi_{t+1}(\mu_t^{(i)} - \mu_0)$$

$$P_{t+1|t}^{(i)} = \Phi_{t+1} P_t^{(i)} \Phi_{t+1}^\top + Q_{t+1}$$

The conditional mean and covariance follow the Rauch-Tung-Striebel (RTS) backward sampling equations for each particle $i = 1, \dots, N$:

$$m_{t|t+1}^{(i)} = \mu_t^{(i)} + J_t^{(i)} (X_{t+1}^* - \mu_{t+1|t}^{(i)})$$

$$\Sigma_{t|t+1}^{(i)} = \Sigma_t^{(i)} - J_t^{(i)} P_{t+1|t}^{(i)} J_t^{(i)\top}$$

$$J_t^{(i)} = \Sigma_t^{(i)} \Phi_{t+1}^\top P_{t+1|t}^{(i)-1}$$

Sampling: 1. Sample component index: $I_t \sim \text{Categorical}(w_{t|t+1}^{(1)}, \dots, w_{t|t+1}^{(N)})$ 2. Sample state: $X_t^* \sim \mathcal{N}(m_{t|t+1}^{(I_t)}, \Sigma_{t|t+1}^{(I_t)})$

Appendix

2.1 Preserving Kronecker Structure

Assuming $\Phi_t = \phi_t \otimes I_H$ where $\phi_t \in \mathbb{R}^{F \times F}$ and $I_H \in \mathbb{R}^{H \times H}$, we can decompose the predictive covariance into a Kronecker product structure as follows:

$$\begin{aligned}
P_{t|t-1} &= \Phi_t \Sigma_{t-1} \Phi_t^\top + Q_t \\
&= \Phi_t (\Gamma_{t-1} \otimes B) \Phi_t^\top + (\Gamma_0 \otimes B) - \Phi_t (\Gamma_0 \otimes B) \Phi_t^\top \\
&= (\phi_t \otimes I_H) (\Gamma_{t-1} \otimes B) (\phi_t^\top \otimes I_H) + (\Gamma_0 \otimes B) - (\phi_t \otimes I_H) (\Gamma_0 \otimes B) (\phi_t^\top \otimes I_H) \\
&= (\phi_t \Gamma_{t-1} \phi_t^\top) \otimes (I_H B I_H) + \Gamma_0 \otimes B - (\phi_t \Gamma_0 \phi_t^\top) \otimes (I_H B I_H) & \text{(mixed-product)} \\
&= (\phi_t \Gamma_{t-1} \phi_t^\top) \otimes B + \Gamma_0 \otimes B - (\phi_t \Gamma_0 \phi_t^\top) \otimes B \\
&= (\phi_t \Gamma_{t-1} \phi_t^\top + \Gamma_0 - \phi_t \Gamma_0 \phi_t^\top) \otimes B & \text{(distributivity)} \\
&= \Gamma_t^- \otimes B
\end{aligned}$$

where $\Gamma_t^- = \phi_t \Gamma_{t-1} \phi_t^\top + \Gamma_0 - \phi_t \Gamma_0 \phi_t^\top$. We also used the **mixed-product property** of Kronecker products: $(A \otimes B)(C \otimes D) = AC \otimes BD$ and the **distributivity property**: $A \otimes C + B \otimes C = (A + B) \otimes C$. Since the marginal distribution of latent states is conditionally Gaussian, we can decompose the predictive covariance into a block matrix form as follows:

$$P_{t|t-1} = \begin{bmatrix} P_{t|t-1}^{EE} & P_{t|t-1}^{ER} \\ P_{t|t-1}^{RE} & P_{t|t-1}^{RR} \end{bmatrix} = \begin{bmatrix} \Gamma_t^{EE,-} & \Gamma_t^{ER,-} \\ \Gamma_t^{RE,-} & \Gamma_t^{RR,-} \end{bmatrix} \otimes B$$

Following our Kalman update, we can decompose the Kalman gain as

$$\begin{aligned}
K_t &= P_{t|t-1}^{RE} (P_{t|t-1}^{EE})^{-1} \\
&= (\Gamma_t^{RE,-} \otimes B) (\Gamma_t^{EE,-} \otimes B)^{-1} \\
&= (\Gamma_t^{RE,-} \otimes B) ((\Gamma_t^{EE,-})^{-1} \otimes B^{-1}) & \text{(inverse property)} \\
&= \Gamma_t^{RE,-} (\Gamma_t^{EE,-})^{-1} \otimes I_P & \text{(mixed-product)}
\end{aligned}$$

where $\Gamma_t^{RE,-} = \Gamma_t^{RR,-} - \Gamma_t^{RE,-} (\Gamma_t^{EE,-})^{-1} \Gamma_t^{ER,-}$. We also used the **inverse property** of Kronecker products: $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

The filtering distribution for the full state conditional on the observed data $y_{1:t}$ can be represented as

$$p(x_t | y_{1:t}) \approx \sum_{i=1}^N w_t^{(i)} \mathcal{N} \left(\begin{pmatrix} \mu_t^{E,(i)} \\ m_t^{R|E,(i)} \end{pmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & P_t^{RR|E} \end{bmatrix} \right)$$

where

- $P_t^{EE|E} = P_t^{EE} - P_t^{EE} (P_t^{EE})^{-1} P_t^{EE} = 0$
- $P_t^{ER|E} = P_t^{ER} - P_t^{EE} (P_t^{EE})^{-1} P_t^{ER} = 0$, and
- $P_t^{RE|E} = P_t^{RE} - P_t^{RE} (P_t^{EE})^{-1} P_t^{EE} = 0$. The covariance matrix $\begin{bmatrix} 0 & 0 \\ 0 & P_t^{RR|E} \end{bmatrix} \in \mathbb{R}^{HP \times HP}$ where $P_t^{RR|E} \in \mathbb{R}^{(F-H)P \times (F-H)P}$.

The conditions for the Kronecker product structure to be preserved are:

1. The initial covariance matrix Σ_0 must have a Kronecker product structure, i.e., $\Sigma_0 = \Gamma_0 \otimes B$.
2. $\Gamma_t^{EE,-}$ must be positive definite (hence invertible).

3. B must be positive definite (hence invertible) for the Kalman gain to preserve Kronecker structure for the latent states involved in an observation.

3.1 Preserving Kronecker Structure in Smoothing

Assuming the filtering covariance maintains the Kronecker structure $\Sigma_t^{(i)} = \Gamma_t \otimes B$ and the predictive covariance is $P_{t+1|t}^{(i)} = \Gamma_{t+1}^- \otimes B$, we can decompose the backward sampling gain following the Kronecker product structure:

$$\begin{aligned} J_t^{(i)} &= \Sigma_t^{(i)} \Phi_{t+1}^\top P_{t+1|t}^{(i),-1} \\ &= (\Gamma_t \otimes B)(\phi_{t+1}^\top \otimes I_P)(\Gamma_{t+1}^- \otimes B)^{-1} \\ &= (\Gamma_t \otimes B)(\phi_{t+1}^\top \otimes I_P)((\Gamma_{t+1}^-)^{-1} \otimes B^{-1}) \quad (\text{inverse property}) \\ &= \Gamma_t \phi_{t+1}^\top (\Gamma_{t+1}^-)^{-1} \otimes I_P \quad (\text{mixed-product}) \end{aligned}$$

where we used the **mixed-product property** $(A \otimes B)(C \otimes D) = AC \otimes BD$ and the **inverse property** $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

The conditional covariance for backward sampling can be decomposed as:

$$\begin{aligned} \Sigma_{t|t+1}^{(i)} &= \Sigma_t^{(i)} - J_t^{(i)} P_{t+1|t}^{(i)} J_t^{(i)\top} \\ &= (\Gamma_t \otimes B) - (\Gamma_t \phi_{t+1}^\top (\Gamma_{t+1}^-)^{-1} \otimes I_P)(\Gamma_{t+1}^- \otimes B)(\Gamma_{t+1}^-)^{-1} \phi_{t+1} \Gamma_t \otimes I_P \\ &= (\Gamma_t \otimes B) - (\Gamma_t \phi_{t+1}^\top (\Gamma_{t+1}^-)^{-1} \Gamma_{t+1}^- (\Gamma_{t+1}^-)^{-1} \phi_{t+1} \Gamma_t) \otimes B \\ &= (\Gamma_t - \Gamma_t \phi_{t+1}^\top (\Gamma_{t+1}^-)^{-1} \phi_{t+1} \Gamma_t) \otimes B \\ &= \Gamma_{t|t+1} \otimes B \end{aligned}$$

where $\Gamma_{t|t+1} = \Gamma_t - \Gamma_t \phi_{t+1}^\top (\Gamma_{t+1}^-)^{-1} \phi_{t+1} \Gamma_t$, which is also the Schur complement of Γ_{t+1}^- .

The conditions for the Kronecker product structure to be preserved in smoothing are:

1. The filtering covariance $\Sigma_t^{(i)}$ must have Kronecker structure, i.e., $\Sigma_t^{(i)} = \Gamma_t \otimes B$.
2. The predictive covariance Γ_{t+1}^- must be positive definite (hence invertible).
3. B must be positive definite (hence invertible) for the backward gain to preserve Kronecker structure.

4 Simulation Study

We evaluate the RB-PF with Kronecker structure on a simulated dataset. Latent states are generated from a correlated OU-process on $F = 10$ dimensions with $P = 2$ dimensions for the latent states involved in an observation, and $P = 2$ for the likelihood function.

$$X_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$$

where $\Sigma_0 = \Gamma_0 \otimes B$ with $\Gamma_0 \in \mathbb{R}^{10 \times 10}$ and $B \in \mathbb{R}^{2 \times 2}$. The state transition is governed by the OU-process defined as

$$X_t = \mu_0 + \Phi_t(X_{t-1} - \mu_0) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t)$$

where $Q_t = \Sigma_0 - \Phi_t \Sigma_0 \Phi_t^\top$ and $\Phi_t = \phi_t \otimes I_P$ with $\phi_t = \exp(-\kappa \Delta t) \cdot I_F$. The likelihood function is defined as a non-linear function of inputs X_t^E . For simplicity, we utilize a bivariate Gaussian likelihood function for the observed data $y_t \in \mathbb{R}^2$:

$$G_t(y_t \mid x_t^i, x_t^j) = \frac{1}{2\pi\sqrt{|R|}} \exp\left(-\frac{1}{2}(y_t - \mu)^\top R^{-1}(y_t - \mu)\right)$$

where $\mu = [x_t^{1,i} - x_t^{2,j}, x_t^{1,j} - x_t^{2,i}]^\top$ and R is the observation noise covariance.

We aim to estimate the following parameters: κ , R , μ_0 and Σ_t , which decomposes into Γ_t and B .

5 Counterexample: Kronecker product covariance structure with time-varying B_t and time-varying Γ_t

7. Obtain new Σ_t with Kronecker structure

1.

$$\begin{aligned} P_{t|t-1} &= \Phi_t \Sigma_{t-1} \Phi_t^\top + \Sigma_0 - \Phi_t \Sigma_0 \Phi_t^\top \\ &= \Phi_t (\Gamma_{t-1} \otimes B_{t-1}) \Phi_t^\top + (\Gamma_0 \otimes B_0) - \Phi_t (\Gamma_0 \otimes B_0) \Phi_t^\top \end{aligned}$$

2. $\Gamma_t = \begin{bmatrix} \Gamma_t^{EE} & \Gamma_t^{ER} \\ \Gamma_t^{RE} & \Gamma_t^{RR} \end{bmatrix}$ and assume $B_t^{EE} = B_t^{RR} = B_t$

3. (NOT COMPLETED)

$$\begin{aligned} K_t &= P_t^{RE} (P_t^{EE})^{-1} \\ &= [\phi_t^{RE} (\Gamma_t^{RE} \otimes B_t) \phi_t^{RE} + (\Gamma_0^{RE} \otimes B_0) (I - \phi_t^{RE} \phi_t^{RE\top})] [(\Gamma_t^{EE} \otimes B_t)^{-1}] \end{aligned}$$